Started on	Friday, 5 December 2025, 9:29 PM
State	Finished
Completed on	Friday, 5 December 2025, 9:38 PM
Time taken	9 mins 24 secs
Marks	25.00/25.00
Grade	**100.00** out of 100.00

Question 1

Complete

Mark 1.00 out of 1.00

How to run multiple versions of an app simultaneously in Kubernetes?

- a. Deployment
- b. ConfigMap
- c. ReplicaSet
- d. StatefulSet

Question 2

Complete

Mark 1.00 out of 1.00

In Horizontal Pod Autoscaler (HPA), which metric is used by default?

- a. Memory usage
- b. Disk usage
- c. Network I/O
- d. CPU utilization

Question 3

Complete

Mark 1.00 out of 1.00

In Kubernetes, which component is responsible for assigning Pods to Nodes?

- a. kube-scheduler
- b. kubelet
- c. kube-apiserver
- d. kube-controller-manager

Question 4

Complete

Mark 1.00 out of 1.00

Kube-Proxy operates at which OSI layer in IPVS mode?

- a. Layer 2
- b. Layer 3
- c. Layer 7
- d. Layer 4

Question 5

Complete

Mark 1.00 out of 1.00

What does a DaemonSet ensure?

- a. One Pod across the entire cluster
- b. Pods run only on master nodes
- c. One Pod on each worker node
- d. Pods only run when manually scheduled

Question 6

Complete

Mark 1.00 out of 1.00

What does ETCD primarily store?

- a. Pod metrics
- b. Node logs
- c. Cluster state
- d. Container images

Question 7

Complete

Mark 1.00 out of 1.00

What does the kube-controller-manager do?

- a. Runs controllers like Deployment, Node, Replication
- b. Stores cluster state
- c. Routes API traffic
- d. Schedules Pods

Question 8

Complete

Mark 1.00 out of 1.00

What does the status CrashLoopBackOff indicate?

- a. Node is out of memory
- b. Container repeatedly crashes and restarts
- c. ImagePullBackOff due to missing image
- d. Liveness probe disabled

Question 9

Complete

Mark 1.00 out of 1.00

What happens when a Node remains NotReady for a long time?

- a. Pods on the node may be evicted and scheduled elsewhere
- b. All pods are deleted immediately
- c. Node is automatically drained
- d. Scheduler stops working

Question 10

Complete

Mark 1.00 out of 1.00

What happens when Pod affinity rules cannot be satisfied?

- a. Pod runs with lower priority
- b. Scheduler rejects the Pod permanently
- c. Pod schedules on any available node
- d. Pod stays in Pending state

Question 11

Complete

Mark 1.00 out of 1.00

What is the default behavior when a Deployment's Pod template is updated?

- a. A new ReplicaSet is created and old Pods are gradually replaced
- b. All old Pods are immediately deleted
- c. The existing ReplicaSet is patched in place
- d. Pods restart automatically with no ReplicaSet change

Question 12

Complete

Mark 1.00 out of 1.00

What is the default restart policy for Pods in a Deployment?

- a. Always
- b. OnFailure
- c. Manual
- d. Never

Question 13

Complete

Mark 1.00 out of 1.00

What is the role of the kubelet?

- a. Execute kubectl commands
- b. Enforce Pod quotas
- c. Ensure containers in PodSpecs are running
- d. Store cluster state

Question 14

Complete

Mark 1.00 out of 1.00

Which API group contains Deployments and ReplicaSets?

- a. apps/v1
- b. core/v1
- c. rbac.authorization.k8s.io/v1
- d. networking.k8s.io/v1

Question 15

Complete

Mark 1.00 out of 1.00

Which authentication method is NOT supported by Kubernetes API server?

- a. SSH keys
- b. X.509 Client Certificates
- c. Static token file
- d. OIDC

Question 16

Complete

Mark 1.00 out of 1.00

Which command generates a manifest without applying it?

- a. kubectl manifest --generate
- b. kubectl create --file-only
- c. kubectl dry-run=client -o yaml
- d. kubectl simulate=apply

Question 17

Complete

Mark 1.00 out of 1.00

Which command lists all API resource types in the cluster?

- a. kubectl get resources
- b. kubectl list kinds
- c. kubectl api-resources
- d. kubectl get all --all-namespaces

Question 18

Complete

Mark 1.00 out of 1.00

Which component ensures cross-node Pod-to-Pod networking?

- a. Kubelet
- b. API server
- c. CNI plugin
- d. kube-proxy

Question 19

Complete

Mark 1.00 out of 1.00

Which Kubernetes object provides persistent storage across Pod restarts?

- a. PersistentVolumeClaim
- b. HostPath
- c. ConfigMap
- d. EmptyDir

Question 20

Complete

Mark 1.00 out of 1.00

Which object controls communication rules between Pods at Layer-3/4?

- a. ConfigMap
- b. NetworkPolicy
- c. ServiceAccount
- d. RoleBinding

Question 21

Complete

Mark 1.00 out of 1.00

Which object provides stable identity and persistent storage for replicas?

- a. ReplicaSet
- b. DaemonSet
- c. StatefulSet
- d. Deployment

Question 22

Complete

Mark 1.00 out of 1.00

Which policy prevents voluntary disruptions of Pods?

- a. RBAC Policy
- b. ResourceQuota
- c. NetworkPolicy
- d. PodDisruptionBudget

Question 23

Complete

Mark 1.00 out of 1.00

Which Service type creates an externally accessible load-balanced IP?

- a. Headless
- b. NodePort
- c. ClusterIP
- d. LoadBalancer

Question 24

Complete

Mark 1.00 out of 1.00

Which Service type provides DNS but no cluster IP or load balancer?

- a. Ingress
- b. Headless Service (ClusterIP: None)
- c. ExternalName
- d. NodePort

Question 25

Complete

Mark 1.00 out of 1.00

Which volume type enables dynamic provisioning?

- a. StorageClass-backed PersistentVolume
- b. HostPath
- c. EmptyDir
- d. Manually created PersistentVolume